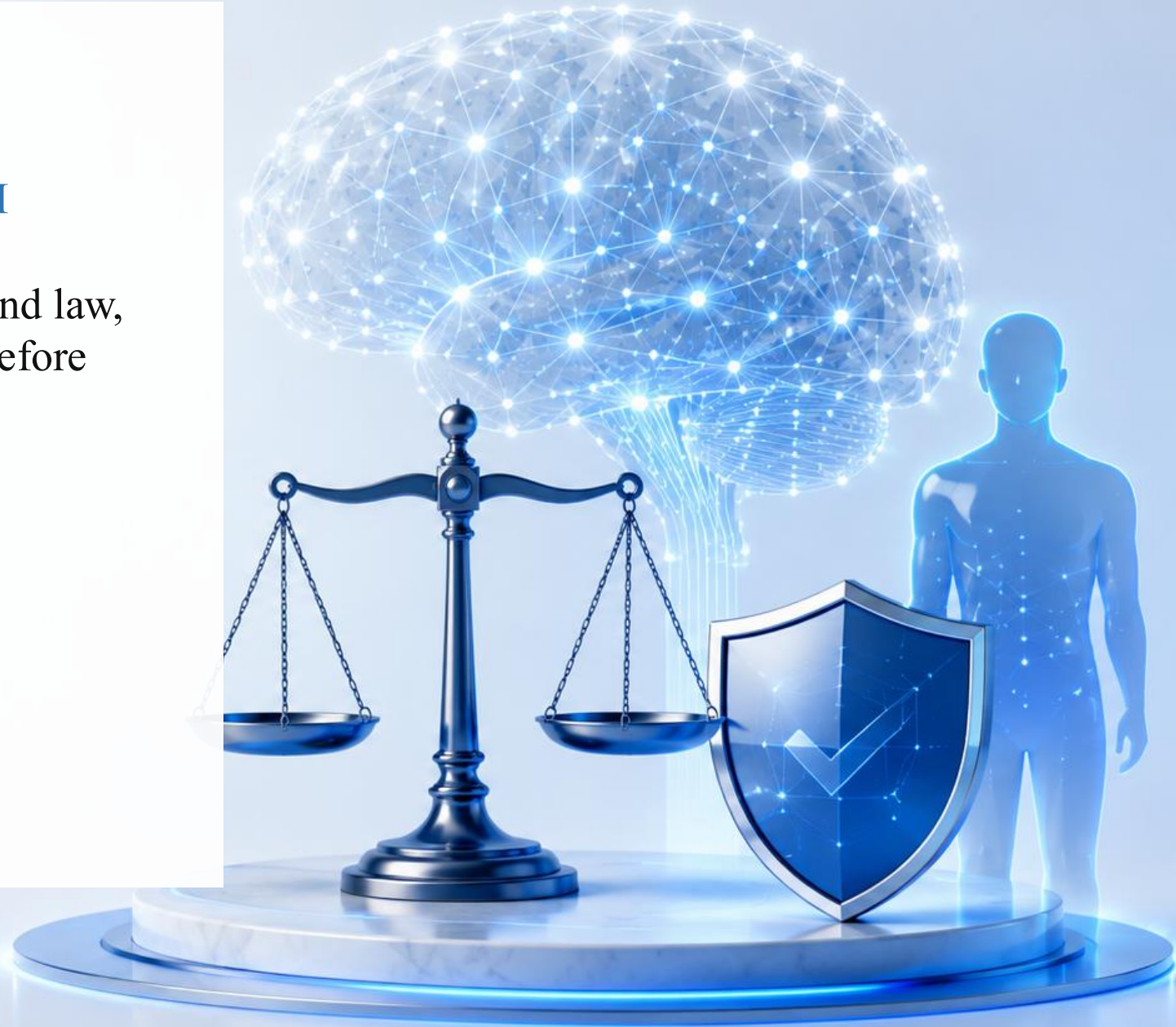


Unit 1 Lecture 2: AI Ethics

TCS349: Responsible and Explainable AI

Today we will understand ethics, morality and law, and why AI systems need ethical thinking before deployment.

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Why are we studying AI Ethics?

AI systems increasingly make decisions that affect people.
A model may be accurate but still harmful, biased or unfair.
Ethical AI asks: should we build it, how should we build it and who is responsible?
Responsible AI begins with ethical thinking, not only coding.

*A good AI system is not only intelligent.
It must also be responsible, fair and safe.*



Opening question: Can an AI system be useful but still unethical? Give one example.

What is Ethics?

Ethics is the study of what is right, wrong, fair and responsible. It helps us decide what we should do, not only what we can do. In AI, ethics guides design, data collection, model training and deployment.

Ethical questions usually involve people, harm, rights, responsibility and trust.

Technical question: Can we build it?
Ethical question: Should we build it?



Teaching cue: Ask students whether marks-prediction AI for students is helpful, harmful or both.

Ethics, Morality and Law

Concept	Meaning	AI Example
Ethics	Systematic principles for right and responsible action	Should a college use AI to monitor student attention?
Morality	Personal or social belief about right and wrong	A teacher may feel constant monitoring is disrespectful.
Law	Rules enforced by government or authority	Data protection laws may restrict student data collection.

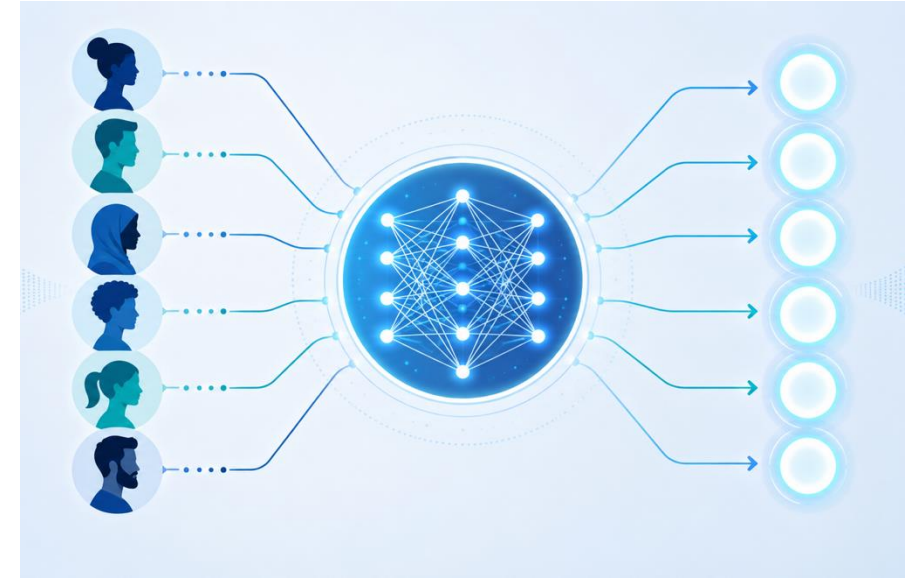
Important: Something can be legal but still ethically questionable.

Why AI Ethics is Different from General Software Ethics

AI systems learn patterns from data, so hidden bias can enter silently. Predictions may affect real people: loans, jobs, healthcare, policing, education.

Many AI models are difficult to explain, especially deep learning models.

AI systems may scale quickly and harm thousands of people at once. Responsibility is shared among developers, organizations, users and regulators.



Teaching point

AI ethics is not a separate chapter after coding. It must be considered throughout the AI lifecycle.

Core Principles of AI Ethics

Fairness

AI should not systematically disadvantage a person or group.

Transparency

People should understand how and why AI is being used.

Accountability

There should be clear responsibility for AI decisions and failures.

Privacy

AI should respect personal data, consent and confidentiality.

Safety

AI should avoid foreseeable harm and work reliably in real conditions.

Human Oversight

Humans should remain involved in sensitive and high-risk decisions.

Quick check: Which principle is most important in healthcare AI? Why?

Common Ethical Dilemmas in AI

Accuracy vs fairness: A model may be accurate overall but unfair for some groups.

Privacy vs personalization: Better recommendations may require more personal data.

Automation vs employment: AI may improve productivity but reduce some jobs.

Security vs surveillance: Monitoring may improve safety but reduce freedom.

Speed vs accountability: Automated decisions are fast, but who explains mistakes?



Key idea

Ethical dilemmas often do not have a perfect answer. We balance values and reduce harm.

Ethical Dilemma: Autonomous Vehicle Decision



Scenario

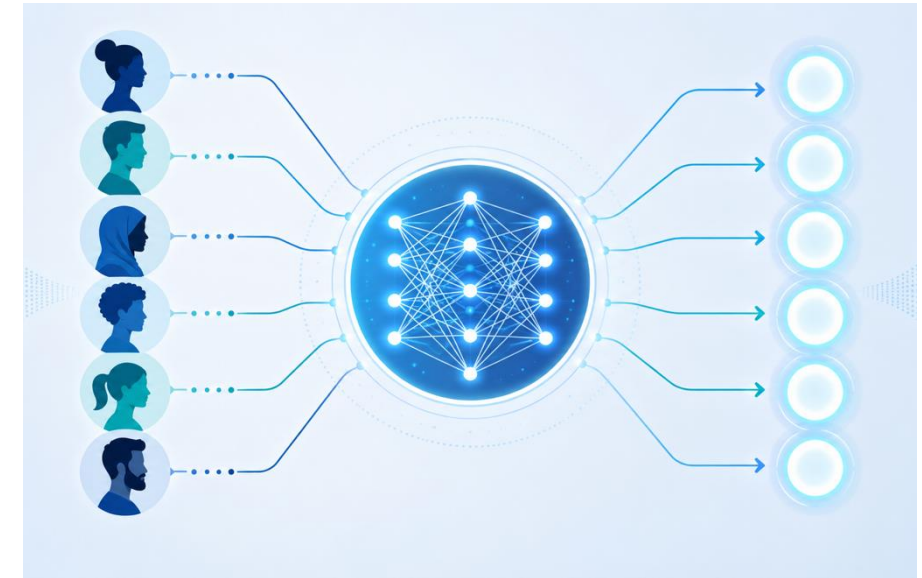
A self-driving car faces an unavoidable accident.
Any action may harm someone.

Who should decide the vehicle behavior?
Developer? Company? Government? User?
Should the car protect passengers first?
How should such decisions be tested and regulated?

Discussion: Should an autonomous vehicle make moral decisions or simply follow traffic rules?

Case Discussion: AI in Hiring

A company uses AI to shortlist resumes from thousands of applicants. The model learns from previous hiring patterns. If past hiring was biased, the AI may repeat or amplify that bias. The system may reject deserving candidates without proper explanation.



Case Discussion: AI in Healthcare

Situation

An AI model predicts disease risk from medical images or patient records.

Benefit

Early diagnosis, faster screening and support for doctors.

Risk

Wrong prediction may delay treatment or create panic.

Should AI give final medical decisions?

How much explanation is required for doctors and patients?

Who is responsible if the AI prediction is wrong?

How should patient data privacy be protected?

Privacy Dilemma: Convenience vs Consent



AI assistants, apps and platforms collect user data to improve services.

More data can improve personalization and accuracy.

But users may not know what data is collected, stored or shared.

Ethical AI requires consent, data minimization and protection.

Question: Does “I agree” always mean informed consent?

A Simple 5-Question Ethical Framework

- 1 Who can be affected by this AI system?
- 2 What data is being used, and was consent taken?
- 3 Can the system create bias, exclusion or unfair harm?
- 4 Can users understand or challenge the decision?
- 5 Who is accountable if something goes wrong?

Use this framework for any AI system before implementation or deployment.